

AnchorIQ

Customer Churn & Retention Analysis

Dataset: telco_churn.csv

Generated: October 03, 2026 - 10:18 | Powered by AnchorIQ v2.0

7,043
Records

21
Features

Churn
Target

0
Fixed Nulls

0
Dupes Cleared

1. EXECUTIVE SUMMARY & AI INSIGHTS

■ Executive Summary

Your customer base of 7,043 subscribers has an overall churn rate of 26.5%. This is high, but manageable. The data shows that churn is not random; it is heavily concentrated in specific groups. By focusing on customers with short tenure, month-to-month contracts, and fiber optic internet, you can address the majority of your risk. A predictive model has been built that identifies the highest-risk customers with strong accuracy, allowing you to target your retention efforts efficiently rather than spreading resources thin.

■ Key Findings

- **Short Tenure is the Biggest Risk:**** Customers with 0–9 months of tenure have a churn rate of 49.8%, nearly double the overall average. This group represents 26% of your customers.
- **Contract Type Matters:**** Month-to-month customers churn at a rate of 42.7%, compared to the 26.5% baseline. This is your largest at-risk segment, comprising 55% of all records.
- **Fiber Optic Users Are Vulnerable:**** Customers on Fiber optic internet churn at 41.9%. This is a significant portion of your base (44%) and suggests that service quality or pricing for this specific tier may be driving dissatisfaction.
- **Low Total Spend Correlates with Churn:**** Customers with total charges between \$18.80 and \$402.20 churn at 43.4%. This likely overlaps with the short-tenure group, indicating that customers who haven't invested much in your service are less loyal.
- **Protective Factor:**** Customers without internet service have a very low churn rate of only 7.4%. This suggests that phone-only customers are far more stable than those using data services.

■ Recommendations

- **Target the Top 10% Highest-Risk Customers****
 - **Why:**** The model identifies 704 customers as highest risk. This small group contains 29% of all actual churn cases.
 - **Impact:**** By contacting only these 704 people, you reach nearly a third of your churn problem while touching only 10% of your customer base. This is the most efficient use of your retention budget.
- **Offer Retention Incentives to Month-to-Month Customers****
 - **Why:**** This group has a 42.7% churn rate.
 - **Impact:**** If you can reduce the churn rate in this group by just 10% (relative), you would save approximately 166 customers. Consider offering a discount for switching to a 12-month or 24-month contract.

3. **Review Fiber Optic Service Value**

- **Why:** Fiber optic users churn at 41.9%.
- **Impact:** A 10% relative reduction in churn for this group would save about 130 customers. Investigate if there are specific service issues or if the price point is too high for the value perceived.

4. **Engage New Customers (0–9 Months Tenure)**

- **Why:** This group has the highest churn rate at 49.8%.
- **Impact:** A 10% relative reduction here saves about 92 customers. Implement a "welcome" retention program or check-in call during the first few months to ensure they are satisfied.

■ ■ **Risks & Caveats**

- **Model Limitations:** The model has an AUC of 0.84, which is good but not perfect. It correctly flags 73% of real churn cases, but 44% of its flags are false positives (it predicts churn for customers who stay). Do not rely on it as the sole decision-maker.
- **Assumptions in Impact Estimates:** The savings estimates (e.g., "166 fewer churn cases") assume a 10% relative reduction in churn rates. Actual results will depend on the effectiveness of your specific retention offers.
- **Data Quality:** There are outliers in the data, such as one record with total charges of \$

2. DATASET OVERVIEW & QUALITY AUDIT

Audit Metric	Value	Status
Total Records	7,043	Validated
Total Feature Count	21	Validated
Numerical Columns	3	Active
Categorical Columns	18	Active
Target Column	Churn	Identified
Missing Cells Imputed	0	Resolved
Duplicate Rows Purged	0	Resolved
Data Quality Index	100.0%	Optimal

Preprocessing Pipeline Execution:

- Converted 'TotalCharges' to numeric
- Filled 11 nulls in 'TotalCharges' with median (1397.47)

3. MACHINE LEARNING BENCHMARKS

Problem Formulation: **Classification** | Best Champion: **Logistic Regression** | Training: **5,634** samples | Evaluation: **1,409** samples | Engineered Features: **19**

Model Candidate	AUC Score	Validation Status
Champion: Logistic Regression	AUC = 0.8414	Top Performer
Logistic Regression	AUC = 0.8449	Linear
Random Forest	AUC = 0.8423	Ensemble
XGBoost Classifier	AUC = 0.8408	Gradient Boosted
Accuracy	73.74%	Overall Match
Cross-Validation	0.8449 ± 0.006	5-Fold CV

Key Drivers & Influential Signals:

Rank	Feature Variable	Importance Weight
1	tenure	0.4783
2	InternetService	0.1578
3	Contract	0.1155
4	MonthlyCharges	0.1076
5	TotalCharges	0.0386
6	StreamingMovies	0.0181
7	StreamingTV	0.0171
8	OnlineSecurity	0.0140

4. EXPLORATORY DATA ANALYSIS (EDA CHARTS)

Figure 1: Feature Distributions

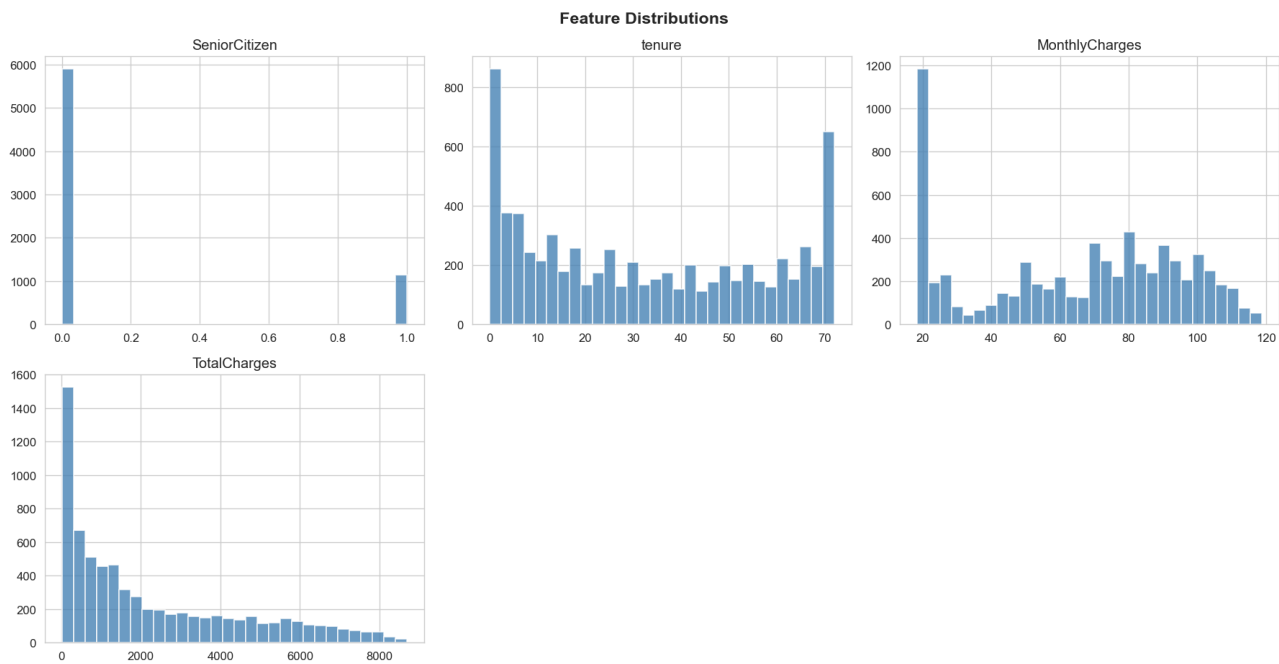


Figure 2: Feature Correlation Heatmap

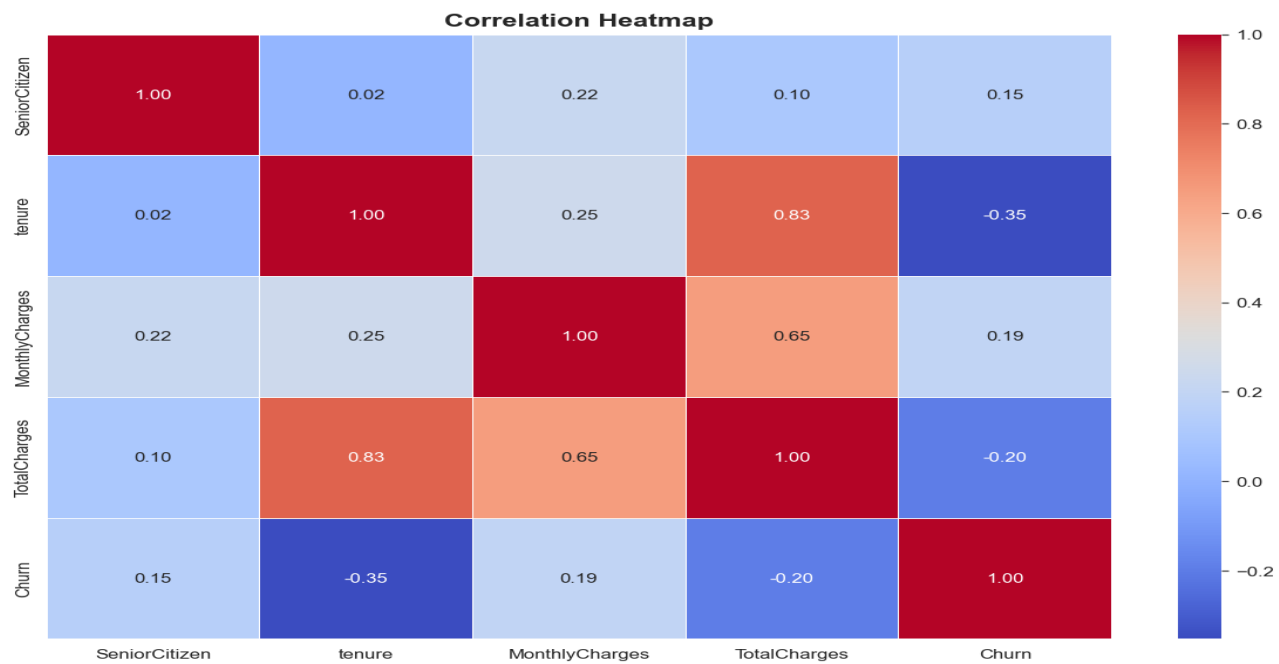


Figure 3: Target Variable Distribution

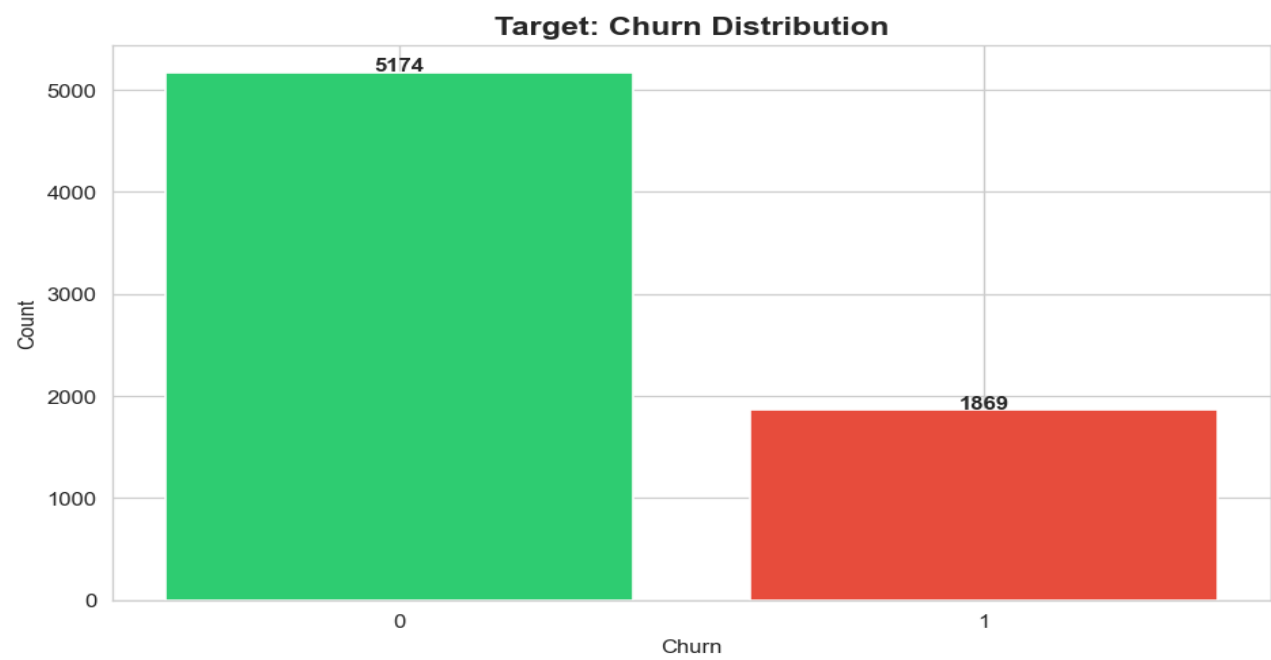


Figure 4: Categorical Feature Impact

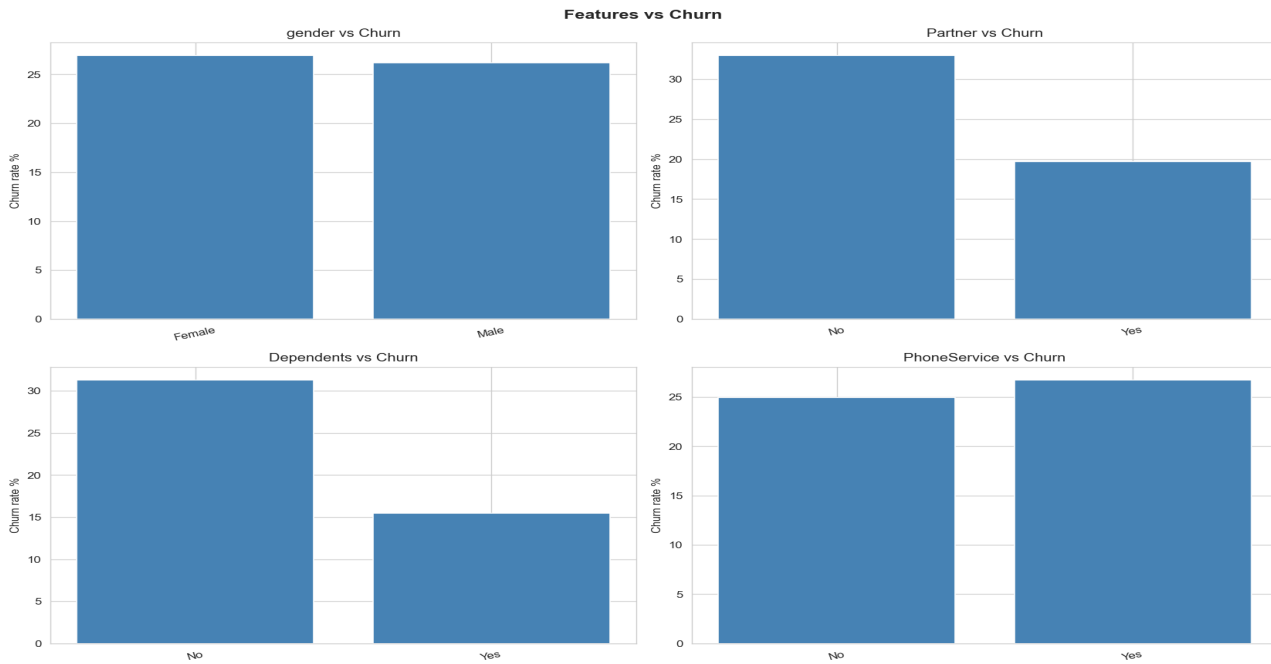
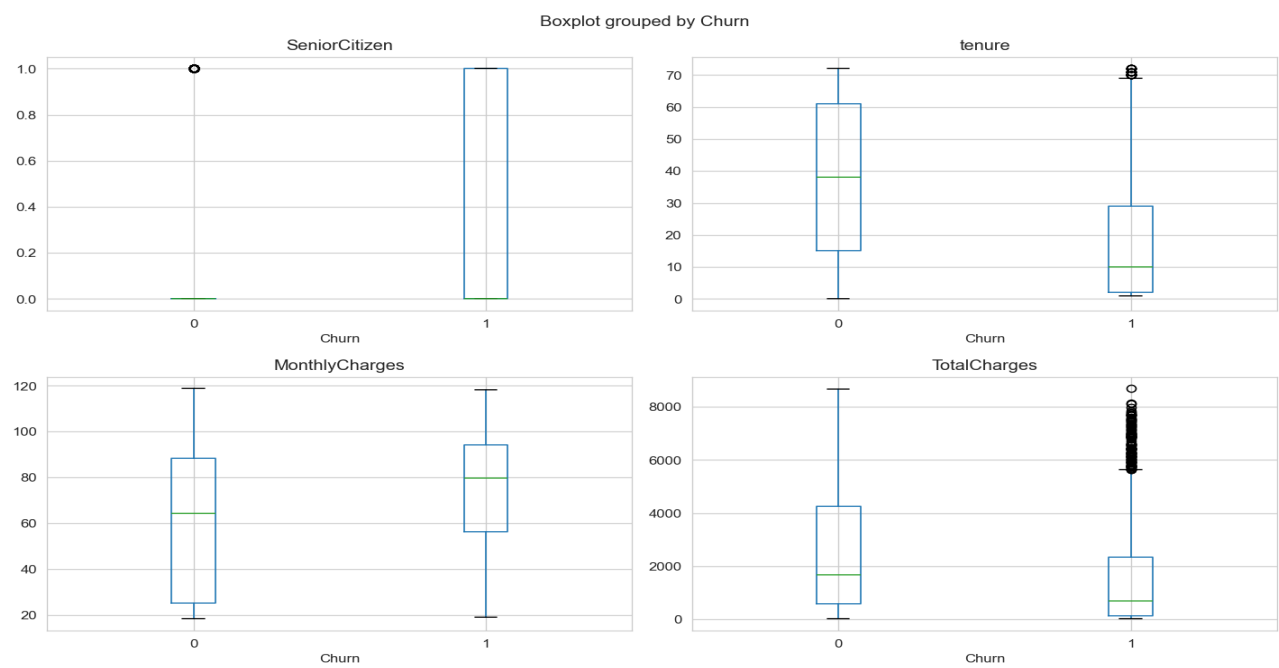


Figure 5: Numerical Distributions by Target



5. FEATURE STATISTICAL SUMMARY

Feature	Mean	Median	Std Dev	Min	Max	Skewness
SeniorCitizen	0.16	0.0	0.37	0.0	1.0	1.83
tenure	32.37	29.0	24.56	0.0	72.0	0.24
MonthlyCharges	64.76	70.35	30.09	18.25	118.75	-0.22